Project Proposal

1. Purpose of This Proposal

Before you begin building your Comprehensive Networking Project in Cisco Packet Tracer, complete this proposal and submit it for instructor approval. It documents your project's scope and design — title, scenario, objectives, functional requirements, network design, and timeline — and, in the final sections, requires you to explicitly show how your project design engages with the Complex Engineering Problem (CEP) and Complex Engineering Activity (CEA) attributes defined by the Board of Accreditation for Engineering and Technical Education (BAETE), in line with the Washington Accord graduate-attribute framework.

A project of this scale will not necessarily satisfy every CEP/CEA attribute to the same depth as a capstone or thesis project — that is expected. What matters is that you honestly identify which attributes your specific design choices engage with, and justify your answer with reference to your own scenario and network design, not generic statements.

2. Project Title

University Campus Network Using Cisco Packet Tracer

Scenario: Smart University Campus Network for Premier University, Chittagong (proposed simulation) | Platform: Cisco Packet Tracer [insert installed version] | Category: Comprehensive University Campus Network Design and Simulation

3. Project Description / Scenario

Describe your chosen smart city/town/region scenario in one paragraph, as required by the project brief: what zones/sites does it include, and why does it need multi-zone routing, VLAN segmentation, VLSM/IPv6 addressing, wireless connectivity, ACL/NAT security, and an IoT-driven smart-city function?

This project proposes the design and simulation, in Cisco Packet Tracer, of a smart university campus network modelled on Premier University, Chittagong. The simulated campus is organised into two building blocks containing nine functional segments: an Academic & Library block (computer laboratories, the library and research area, student Wi-Fi and a smart-campus IoT segment) and an Administration & Data Center block (administration and registrar office, faculty and staff, the server room and network management), both connected to the Internet through a WAN edge. These segments are proposed for simulation only and do not describe the university's actual infrastructure. The stakeholders have different and partly conflicting needs: students require broad but restricted access, the registrar's office holds confidential academic and financial records, faculty need both teaching and administrative services, researchers need IPv6-capable connectivity, facilities staff need dependable automation, and IT administrators need secure, centralised management. A single flat network cannot meet these needs, so the design uses multiple routed segments — an ISR4331 edge router and two Catalyst 3650 multilayer distribution switches joined by point-to-point links running OSPF, which selects paths automatically and reroutes traffic if a link fails. Within each block, VLANs separate user groups and switched virtual interfaces (SVIs) perform inter-VLAN routing. A VLSM plan drawn from 10.10.0.0/16 sizes every subnet to its host requirement with room for growth, while dual-stack IPv6 on the library/research and server segments prepares the campus for IPv6 adoption. WPA2-secured Wi-Fi gives students mobile access; named ACLs keep students out of administrative, management and IoT segments; and NAT/PAT at the edge lets privately addressed hosts reach the Internet while publishing only the campus web portal. Finally, an IoT segment registered to a central controller automates classroom lighting by occupancy and ventilation by temperature, showing how the network can support energy-aware campus operations.

> **Scope adaptation notes for instructor approval.** (1) The template's smart-city/town/region examples are adapted to a smart-campus scenario; "zones" are interpreted as campus functional segments grouped into two routed building blocks, not geographically separate sites joined by WAN circuits. If the brief requires a separate router per zone or serial WAN links, more routers can join the OSPF domain without changing the addressing plan. (2) IPv6 is proposed for internal segments only (VLANs 50 and 60 and link L3); IPv6 Internet access is out of scope because no IPv6 ISP is simulated. (3) Wireless uses WPA2-Personal (AES) as the baseline; WPA2-Enterprise (802.1X/RADIUS) is an optional extension subject to Packet Tracer support. (4) Lab sessions named in the template (Labs 4, 11 and 13) are cited as given in the brief and should be confirmed against the course schedule. (5) OSPFv3/IPv6 routing on the 3650 model and the ACL "established" keyword must be confirmed in the installed Packet Tracer version; fallbacks are given in Section 8.

4. Objectives

List 3–5 specific objectives your project aims to achieve (what your network will let a user/device do, and which networking skills it is designed to demonstrate).

1.	**Build a routed, multi-block campus topology** in Cisco Packet Tracer with one edge router (ISR4331), two multilayer distribution switches (3650-24PS) and five access switches (2960-24TT), in which all three Layer-3 devices form FULL OSPFv2 adjacencies in Area 0 and every permitted segment reaches every other (target: 100% of the planned reachability tests pass).
2.	**Allocate an error-free VLSM IPv4 plan** from 10.10.0.0/16 for nine VLANs and three point-to-point links, with no overlapping subnets and at least 20% spare host capacity in every user VLAN, and deploy dual-stack IPv6 (2001:db8:acad::/48) on the library/research and server VLANs so that a research PC reaches the campus portal over IPv6.
3.	**Enforce an inter-zone security policy** with three named extended ACLs, a VTY access list and NAT/PAT, verified by a test matrix of at least 15 permit/deny cases in which every observed result matches the expected result.
4.	**Provide secure wireless access and centralised services:** student laptops and smartphones join SSID PU-STUDENT with WPA2-PSK (AES) and obtain DHCP leases from a central server through relay, and DNS resolves the portal name portal.pu.internal to its IPv4 and IPv6 addresses.
5.	**Demonstrate a smart-campus IoT function** in which a motion detector controls classroom lighting and a temperature monitor controls ventilation through rules on a central IoT server, with each rule triggering correctly when the simulated environment changes.

5. Functional Requirements (Major Features)

List the major features/functions your network will provide (e.g. inter-zone connectivity, VLAN isolation, secure remote/wireless access, a specific IoT automation, a security policy enforced by an ACL, etc.).

FR1	**Inter-zone routing with redundancy.** R-EDGE, DSW-ACAD and DSW-ADMIN run OSPFv2 (process 10, Area 0) over links L1–L3, and R-EDGE originates a default route. Test: show ip ospf neighbor shows FULL adjacencies; show ip route lists all VLAN subnets on each Layer-3 device; with link L3 shut down, traceroute between blocks reroutes via R-EDGE and connectivity returns after reconvergence.
FR2	**VLAN segmentation and inter-VLAN routing.** VLANs 10, 20, 30, 40, 50, 60, 70, 98 and 99 are created where required; access ports are statically assigned; uplinks are 802.1Q trunks with native VLAN 999 and restricted allowed-VLAN lists; unused ports are shut down in VLAN 998. Test: show vlan brief and show interfaces trunk; hosts in different VLANs communicate only through their SVI gateway.
FR3	**Centralised DHCP and DNS.** SRV-DHCP-DNS (10.10.4.34) serves one pool per user/IoT VLAN, relayed by ip helper-address on each SVI; servers and network devices are statically addressed. DNS resolves portal.pu.internal (A 10.10.4.35, AAAA 2001:db8:acad:60::35) and iot.pu.internal. Test: clients receive the correct address, gateway and DNS server; a lab PC opens http://portal.pu.internal.
FR4	**Secure student wireless access.** AP-STUDENT broadcasts SSID PU-STUDENT with WPA2-PSK (AES), bridged into VLAN 40 (10.10.0.0/23). Test: a laptop and a smartphone with the correct key associate, lease 10.10.0.x addresses and reach the portal and INTERNET-SRV; a client with a wrong key cannot associate.
FR5	**Edge security with ACLs and NAT/PAT.** PAT translates inside user subnets to 203.0.113.2 (IoT and management subnets excluded); static NAT publishes SRV-WEB as 203.0.113.5; ACLs STUDENT-IN, IOT-PROTECT and EDGE-IN enforce the policy in Section 8; device management is SSH-only from the management VLANs. Test: show ip nat translations and ACL match counters; student-to-registrar ping fails while student-to-portal HTTP succeeds; INTERNET-SRV can reach only 203.0.113.5 on TCP 80.
FR6	**IPv6 segment and IoT automation.** VLANs 50 and 60 are dual-stack, with IPv6 routes exchanged over L3 by OSPFv3 (static IPv6 routes as fallback). IoT devices in VLAN 70 register with SRV-IOT (10.10.4.36), which runs the lighting and ventilation rules. Test: a library PC pings and browses 2001:db8:acad:60::35; changing motion and temperature in the simulation switches the light and fan as specified.

6. Proposed Network Components and Their Roles

List every major device, segment, and service you plan to configure. "Type/Role" means, e.g., Core Router, Access/Distribution Switch, Wireless Router/AP, IoT Controller or Sensor, Application Server, or Security Boundary (ACL/NAT).

Device / Segment Name	Type / Role	Key Configuration	Responsibilities
ISP-R and INTERNET-SRV (Internet / WAN edge)	Cisco 2911 router + Server-PT; simulated ISP and public server	ISP-R G0/0 203.0.113.1/29, G0/1 198.51.100.1/24; INTERNET-SRV 198.51.100.10 (HTTP, DNS); no routes to 10.10.0.0/16	Represents the public Internet; demonstrates that only NAT-translated traffic can leave the campus and return.
R-EDGE	Cisco ISR4331; edge router and security boundary (ACL/NAT)	G0/0/0 203.0.113.2/29 (nat outside); G0/0/1 10.10.255.1/30, G0/0/2 10.10.255.5/30 (nat inside); Lo0 10.10.254.1/32; OSPF 10; default route + default-information originate; PAT overload; static NAT 10.10.4.35 ↔ 203.0.113.5; ACL EDGE-IN	Connects the campus to the ISP, translates addresses, filters unsolicited inbound traffic and injects the default route into OSPF.
DSW-ACAD	Catalyst 3650-24PS multilayer switch; distribution and inter-VLAN routing (Academic & Library block)	ip routing; SVIs for VLANs 30, 40, 50, 70, 98; G1/0/24 and G1/0/23 routed (no switchport); OSPFv2 + OSPFv3; ip helper-address 10.10.4.34; ACLs STUDENT-IN (SVI 30/40 in) and IOT-PROTECT (SVI 70 in); AC power supply module added in Packet Tracer	Default gateway and policy point for lab, library, Wi-Fi and IoT users; OSPF neighbour of R-EDGE and DSW-ADMIN.
DSW-ADMIN	Catalyst 3650-24PS multilayer switch; distribution and inter-VLAN routing (Administration & Data Center block)	ip routing; SVIs for VLANs 10, 20, 60, 99; routed G1/0/24 and G1/0/23; OSPFv2 + OSPFv3; DHCP relay; AC power supply module	Gateway for administration, faculty, server and management segments; provides the alternate path for academic-block traffic.
ASW-LAB, ASW-LIB, ASW-IOT, ASW-ADMIN, ASW-DC	Catalyst 2960-24TT access switches	Static access VLAN per port; 802.1Q trunk to its DSW (native VLAN 999, allowed list); PortFast + BPDU Guard on edge ports; port security (maximum 2, sticky) on lab ports; management SVI in VLAN 98 or 99; SSH v2 only	Connect end devices to the correct VLAN and protect edge ports against loops and unauthorised devices.
AP-STUDENT	AccessPoint-PT; wireless access point	SSID PU-STUDENT; WPA2-PSK with AES; connected to an ASW-LIB access port in VLAN 40	Provides student Wi-Fi bridged into VLAN 40 (10.10.0.0/23).
SRV-DHCP-DNS	Server-PT; application server for core services	10.10.4.34/28, gateway 10.10.4.33; DHCP pools for VLANs 10, 20, 30, 40, 50 and 70; DNS zone pu.internal (A and AAAA records)	Assigns IPv4 addresses to clients and IoT devices and resolves campus host names (library PCs also use IPv6 SLAAC).
SRV-WEB	Server-PT; campus web portal	10.10.4.35/28 and 2001:db8:acad:60::35/64; HTTP service; published as 203.0.113.5 by static NAT	Hosts the simulated student/staff portal for internal IPv4/IPv6 users and external users.
SRV-IOT	Server-PT with IoT service; IoT controller (registration server)	10.10.4.36/28; IoT registration service on; automation rules R1 and R2 (Section 8); administrator account [to be set by student]	Registers smart devices and executes condition-based automation; dashboard reachable from NMS-PC and faculty PCs.
SRV-NMS	Server-PT; network management services	10.10.4.37/28; Syslog, NTP and TFTP services	Central logging, time synchronisation and configuration backup for network devices.
IoT devices (VLAN 70)	Sensors: motion detector, temperature monitor. Actuators: smart LED light, ceiling fan	Wired IoT network module; DHCP from 10.10.4.11–10.10.4.30; remote IoT server 10.10.4.36	Report occupancy and temperature and carry out the controller's commands.

Device / Segment Name	Type / Role	Key Configuration	Responsibilities
End devices	PC-PT (lab, library, admin, faculty), Laptop-PT and Smartphone-PT (Wi-Fi), NMS-PC	DHCP clients except NMS-PC (static 10.10.4.90/28); library PCs dual-stack with SLAAC; laptops fitted with a WPC300N wireless module	Represent each user group for reachability, policy and service testing.
Security boundary (logical)	ACL / NAT enforcement points	Inside 10.10.0.0/16, outside 203.0.113.0/29; ACLs on SVIs 30, 40, 70 and R-EDGE G0/0/0; VTY ACL MGMT-ONLY (10.10.4.64/27)	Separates trusted and untrusted zones and enforces least-privilege access between campus segments.

7. Network Topology Diagram

Insert or draw your network topology diagram below. It should show every zone, device, and segment listed above, routed links between zones, VLAN boundaries, the wireless segment, the IoT layer, and where your ACL/NAT security boundary sits.

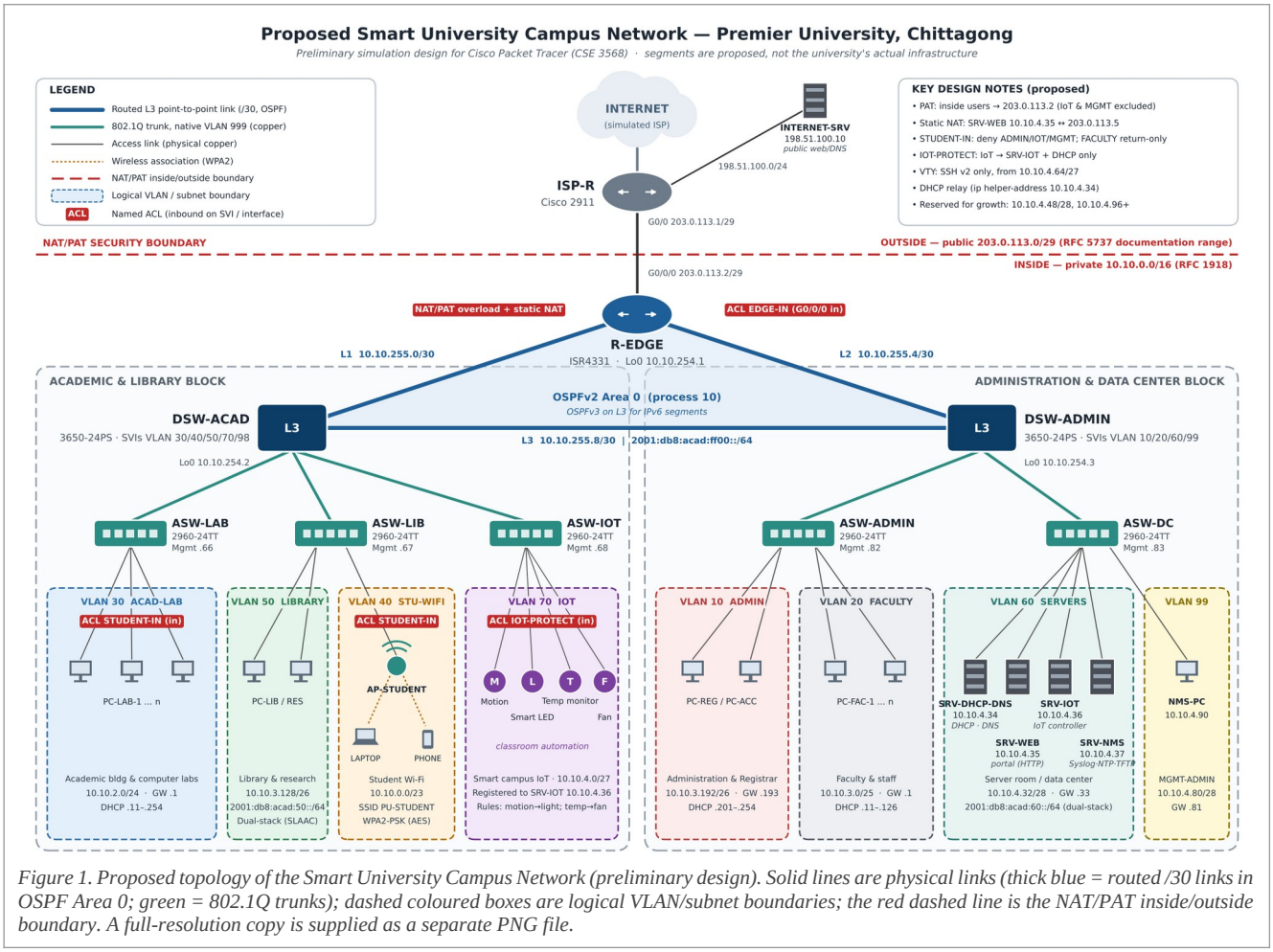


Figure 1. Proposed topology of the Smart University Campus Network (preliminary design). Solid lines are physical links (thick blue = routed /30 links in OSPF Area 0; green = 802.1Q trunks); dashed coloured boxes are logical VLAN/subnet boundaries; the red dashed line is the NAT/PAT inside/outside boundary. A full-resolution copy is supplied as a separate PNG file.

Table 7.1. Preliminary VLAN, IPv4 (VLSM) and IPv6 addressing plan

VLAN	Name / Zone	Gateway device	Hosts needed → prefix (usable)	IPv4 subnet	Default gateway	DHCP scope	IPv6 prefix
10	ADMIN — Administration & Registrar	DSW-ADMIN	40 → /26 (62)	10.10.3.192/26	10.10.3.193	10.10.3.201 – 10.10.3.254	—
20	FACULTY — Faculty & Staff	DSW-ADMIN	100 → /25 (126)	10.10.3.0/25	10.10.3.1	10.10.3.11 – 10.10.3.126	—
30	ACAD-LAB — Academic Bldg & Computer Labs	DSW-ACAD	200 → /24 (254)	10.10.2.0/24	10.10.2.1	10.10.2.11 – 10.10.2.254	—
40	STU-WIFI — Student Wi-Fi	DSW-ACAD	400 → /23 (510)	10.10.0.0/23	10.10.0.1	10.10.0.11 – 10.10.1.254	—
50	LIBRARY — Library & Research	DSW-ACAD	45 → /26 (62)	10.10.3.128/26	10.10.3.129	10.10.3.136 – 10.10.3.190	2001:db8:acad:50::/64 (SLAAC)
60	SERVERS — Server Room / Data Center	DSW-ADMIN	12 → /28 (14)	10.10.4.32/28	10.10.4.33	Static	2001:db8:acad:60::/64 (static)
70	IOT — Smart Campus IoT	DSW-ACAD	20 → /27 (30)	10.10.4.0/27	10.10.4.1	10.10.4.11 – 10.10.4.30	—
98	MGMT-ACAD — Network Mgmt (Academic block)	DSW-ACAD	10 → /28 (14)	10.10.4.64/28	10.10.4.65	Static	—
99	MGMT-ADMIN — Network Mgmt & Security	DSW-ADMIN	10 → /28 (14)	10.10.4.80/28	10.10.4.81	Static	—
—	Reserved (future DMZ / server growth)	—	—	10.10.4.48/28	—	—	—
998 / 999	PARKING (unused ports) / NATIVE (trunks)	All switches	No hosts	No IP subnet	—	—	—
L1	Routed link: R-EDGE G0/0/1 ↔ DSW-ACAD G1/0/24	R-EDGE / DSWs	2 → /30 (2)	10.10.255.0/30	10.10.255.1 / 10.10.255.2	—	—
L2	Routed link: R-EDGE G0/0/2 ↔ DSW-ADMIN G1/0/24	R-EDGE / DSWs	2 → /30 (2)	10.10.255.4/30	10.10.255.5 / 10.10.255.6	—	—
L3	Routed link: DSW-ACAD G1/0/23 ↔ DSW-ADMIN G1/0/23	R-EDGE / DSWs	2 → /30 (2)	10.10.255.8/30	10.10.255.9 / 10.10.255.10	—	2001:db8:acad:ff00::/64 (::1 / ::2)
Lo0	OSPF router-IDs (loopbacks)	R-EDGE / DSW-ACAD / DSW-ADMIN	1 → /32	10.10.254.1 / .2 / .3	—	—	—
WAN	Edge: R-EDGE G0/0/0 ↔ ISP-R G0/0	R-EDGE	public /29	203.0.113.0/29	203.0.113.2 (ISP 203.0.113.1)	—	—
NAT	Static NAT for SRV-WEB; PAT for users	R-EDGE	—	10.10.4.35 ↔ 203.0.113.5	PAT → 203.0.113.2	—	—

Notes: gateways use the first usable address; DHCP scopes exclude low addresses for static devices. Used space is 10.10.0.0–10.10.4.95; 10.10.4.96–10.10.253.255 remains free, and 10.10.4.48/28 is reserved so the server subnet can grow to 10.10.4.32/27. Public addresses come from the RFC 5737 documentation range and IPv6 from the RFC 3849 documentation prefix, because the simulation uses no real provider allocation.

8. Application of Networking Concepts

Explain concretely where and how each core networking concept is applied in your design — name the specific device(s)/zone(s) involved.

Concept	Definition	Where / How Applied in Your Design
Routing	A dynamic routing protocol (OSPF/EIGRP) exchanging reachability information so every zone can reach every other zone.	OSPFv2 (process 10, single **Area 0**) on R-EDGE (router-ID 10.10.254.1), DSW-ACAD (10.10.254.2) and DSW-ADMIN (10.10.254.3). Links L1 (10.10.255.0/30), L2 (10.10.255.4/30) and L3 (10.10.255.8/30) use the point-to-point network type; passive-interface default is applied with only transit links made active, so SVI subnets are advertised without forming adjacencies on user VLANs. R-EDGE holds a static default route to 203.0.113.1 and advertises it with default-information originate. The triangle gives two paths between blocks: L3 is preferred (lower cost) and the path through R-EDGE is the backup. **Rationale:** OSPF is an open standard (RFC 2328) with fast convergence and cost-based path selection; EIGRP was considered but is Cisco-specific, and RIP was rejected for its hop-count metric and slow convergence. **IPv6:** OSPFv3 (RFC 5340) on DSW-ACAD and DSW-ADMIN for VLANs 50, 60 and L3; static IPv6 routes are the fallback if OSPFv3 is unsupported on the 3650 in the installed version.
VLAN Segmentation & Inter-VLAN Routing	Logically separating traffic into VLANs within a zone, then routing between them via a multilayer switch or router-on-a-stick.	Because the inter-block links are routed, VLANs are local to each block and no VLAN spans the core, which keeps each spanning-tree domain small. **DSW-ACAD:** VLAN 30 ACAD-LAB, 40 STU-WIFI, 50 LIBRARY, 70 IOT, 98 MGMT-ACAD. **DSW-ADMIN:** VLAN 10 ADMIN, 20 FACULTY, 60 SERVERS, 99 MGMT-ADMIN. Inter-VLAN routing uses SVIs on the 3650 switches (ip routing) instead of router-on-a-stick, so routed traffic does not hair-pin through the edge router. Trunks to ASW-LAB, ASW-LIB, ASW-IOT, ASW-ADMIN and ASW-DC are 802.1Q with an unused native VLAN 999 and allowed lists (e.g., ASW-LIB: 40, 50, 98); unused access ports are shut down in VLAN 998.
IP Addressing (VLSM & IPv6)	Efficient, hierarchical allocation of address space across zones/VLANs, including at least one IPv6-addressed segment.	**IPv4:** 10.10.0.0/16 (RFC 1918) allocated largest-first by VLSM: VLAN 40 /23, 30 /24, 20 /25, 50 /26, 10 /26, 70 /27, 60 /28, 98 /28 and 99 /28; /30 transit links from 10.10.255.0/24 and /32 loopbacks from 10.10.254.0/24 (full plan in Table 7.1). VLANs 98 and 99 sit in one aligned block (10.10.4.64/27) so a single wildcard (0.0.0.31) matches both in ACLs. **IPv6:** 2001:db8:acad::/48 (RFC 3849 documentation prefix) with one /64 per segment named after the VLAN ID: 2001:db8:acad:50::/64 (library/research, SLAAC), 2001:db8:acad:60::/64 (servers, static) and 2001:db8:acad:ff00::/64 (link L3).
Security (ACL & NAT)	Access control lists enforcing an inter-zone traffic policy, and NAT/PAT translating addresses for outward-facing traffic.	**STUDENT-IN** (extended, inbound on SVIs 30 and 40): permit tcp any 10.10.3.0 0.0.0.127 established; permit icmp any 10.10.3.0 0.0.0.127 echo-reply; deny ip any 10.10.3.0 0.0.0.127 (faculty); deny ip any 10.10.3.192 0.0.0.63 (admin/registrar); deny ip any 10.10.4.0 0.0.0.31 (IoT); deny ip any 10.10.4.64 0.0.0.31 (management); permit ip any any (servers, library, Internet). **IOT-PROTECT** (inbound on SVI 70): permit udp any any eq bootps; permit ip 10.10.4.0 0.0.0.31 host 10.10.4.36; permit icmp 10.10.4.0 0.0.0.31 10.10.4.64 0.0.0.31 echo-reply; deny ip any any. **EDGE-IN** (inbound on R-EDGE G0/0/0): permit tcp any host 203.0.113.5 eq www; permit tcp any any established; permit icmp any any echo-reply; permit udp any eq domain any; deny ip any any. **MGMT-ONLY** (standard, access-class on all VTY lines, SSH v2 only): permit 10.10.4.64 0.0.0.31. **NAT/PAT on R-EDGE:** inside on G0/0/1 and G0/0/2, outside on G0/0/0; ACL NAT-USERS denies 10.10.4.0/27 (IoT) and 10.10.4.64/27 (management) and permits 10.10.0.0/16; ip nat inside source list NAT-USERS interface G0/0/0 overload; ip nat inside source static 10.10.4.35 203.0.113.5.

Concept	Definition	Where / How Applied in Your Design
Wireless Connectivity	A WPA2-secured wireless network (AP/WLC or wireless router) serving end devices or sensors.	AP-STUDENT (AccessPoint-PT) connects to an ASW-LIB access port in VLAN 40 and broadcasts SSID PU-STUDENT with WPA2-PSK and AES encryption (passphrase set by the student and not printed in the report). Laptop-PT (WPC300N module) and Smartphone-PT clients lease from 10.10.0.11–10.10.1.254 through the DSW-ACAD SVI 10.10.0.1, and STUDENT-IN applies to them exactly as to lab PCs. VLAN 40 is sized /23 for high device density. **Optional extension:** WPA2-Enterprise (802.1X) with the AAA/RADIUS service on SRV-NMS, if supported by the installed Packet Tracer version.
IoT / Automation	Sensors/actuators and a controller implementing a condition-based rule for your chosen smart-city function.	Smart devices in VLAN 70 (motion detector, smart LED light, temperature monitor, ceiling fan) lease 10.10.4.11–10.10.4.30 and register with SRV-IOT (10.10.4.36, VLAN 60), reached through DSW-ACAD → L3 → DSW-ADMIN. **Rules on SRV-IOT:** R1 — IF motion detector = On THEN LED light = On; IF motion = Off THEN light = Off (occupancy-based lighting). R2 — IF temperature > 30 °C THEN fan = High; IF temperature < 26 °C THEN fan = Off (the 4 °C gap prevents rapid on/off cycling). Thresholds are initial values to be tuned during testing. IOT-PROTECT limits devices to DHCP and their controller, and NAT-USERS excludes VLAN 70, so IoT devices cannot reach the Internet. The IoT dashboard is opened from NMS-PC or a faculty PC at http://10.10.4.36.

9. Timeline

Fill in your planned dates for each phase. Adjust the tasks below if your own plan differs.

Phase	Planned Dates	Key Task(s)	Deliverable / Milestone
Phase 1: Proposal & Design	**Week 1** *[insert dates]*	Finalize scenario, zone layout, and topology/addressing plan.	Approved proposal; topology diagram and VLSM/IPv6 plan v1.0.
Phase 2: Core Device Configuration	**Week 2** *[insert dates]*	Configure routers/switches and the dynamic routing protocol.	Packet Tracer file v1: devices placed, cabled and named; OSPF adjacencies FULL; default route propagated; SSH enabled.
Phase 3: VLANs & Addressing	**Week 3** *[insert dates]*	Configure VLAN segmentation, inter-VLAN routing, and VLSM/IPv6 addressing.	VLANs, trunks and SVIs verified; DHCP leases in every user VLAN; IPv6 ping between VLANs 50 and 60.
Phase 4: Security & Wireless	**Week 4** *[insert dates]*	Configure ACLs, NAT, and the wireless network.	ACL/NAT test matrix (≥ 15 cases) passed; wireless clients associated and leased.
Phase 5: IoT Integration & Testing	**Week 5** *[insert dates]*	Integrate IoT devices/automation and test all required scenarios.	IoT rules R1–R2 triggered; L3 failover test and full test log recorded.
Phase 6: Report Writing & Submission	**Week 6** *[insert dates]*	Write the Final Report, take screenshots, and submit all deliverables.	Final Report, .pkt file, screenshots and configuration exports submitted.

10. Knowledge Profile (KP) Mapping

Knowledge Profile (K1–K8) describes the range of knowledge underlying complex engineering problem-solving. It also underpins CEP attribute P1 (Depth of Knowledge Required), which draws on K3, K4, K5, K6, and/or K8 in particular. For each area, indicate whether your project design draws on it, and briefly say how.

#	Knowledge Area	Definition (BAETE) & Guiding Question	Your Response
K1	**Natural Sciences**	Theory-based understanding of the natural sciences relevant to the discipline. Often Not Applicable for a networking project — state clearly if no natural-science principles apply, or note if your IoT sensors model a physical quantity (e.g. light, temperature, motion).	Applies? ☐ Yes ☐ Partially ☒ **No** **How:** No natural-science theory is applied in designing or analysing this network. The temperature monitor and motion detector stand for physical quantities (ambient temperature in °C and occupancy), but in Packet Tracer these are simulated values set in the environment panel; no heat-transfer, optical or radio-propagation model is analysed, and Wi-Fi coverage is abstracted by the simulator.
K2	**Mathematics**	Mathematics, numerical analysis, statistics, and formal computer/information science used for analysis and modelling. e.g. binary/subnet-mask arithmetic and VLSM calculations used to derive your addressing plan, or threshold values used in your IoT automation rule.	Applies? ☒ **Yes** ☐ Partially ☐ No **How:** Binary and subnet-mask arithmetic derives the VLSM plan (e.g., 400 hosts → $2^9 - 2 = 510$ usable → /23; 20 hosts → /27), with overlap and headroom checks (about 21–35% spare in user VLANs); wildcard masks are calculated for ACLs (10.10.4.64/27 → 0.0.0.31); IPv6 prefixes are planned in hexadecimal; OSPF costs are compared for path selection; and threshold/hysteresis values (30 °C on, 26 °C off) define the IoT rule.
K3	**Engineering Fundamentals**	Theory-based engineering fundamentals underlying the discipline — for CSE, core networking/computing principles. e.g. the core networking fundamentals your whole design rests on: the OSI/TCP-IP layers, IP addressing, and switching/routing concepts.	Applies? ☒ **Yes** ☐ Partially ☐ No **How:** The design rests on TCP/IP fundamentals: Layer-2 switching and 802.1Q tagging for VLANs, Layer-3 forwarding and longest-prefix matching for inter-VLAN and inter-block routing, ARP and default gateways, transport-layer ports used in ACLs (TCP 80, UDP 53/67), and the difference between stateless filtering and tracking of established TCP sessions.
K4	**Specialist Engineering Knowledge**	Discipline-specific specialist knowledge and frameworks, including current practice areas — e.g. network design and routing. e.g. the specific techniques this course covers: VLSM/IPv6 addressing, dynamic routing protocols (OSPF/EIGRP), VLANs and inter-VLAN routing, ACLs, NAT, and IoT integration.	Applies? ☒ **Yes** ☐ Partially ☐ No **How:** Course-specific techniques are combined: VLSM and dual-stack IPv6, OSPFv2/OSPFv3 with passive interfaces and default-route origination, SVI-based inter-VLAN routing, trunk hardening (unused native VLAN, allowed lists), named extended ACL placement, PAT and static NAT, WPA2-AES wireless, DHCP relay, and IoT registration-server automation.
K5	**Engineering Design**	Knowledge that supports design in the practice area — e.g. network architecture and topology design. e.g. how you designed your zone layout, chose your addressing scheme, or decided where to place security boundaries.	Applies? ☒ **Yes** ☐ Partially ☐ No **How:** Design knowledge is used to choose a two-block hierarchical topology (collapsed core/distribution plus access) with routed inter-block links, to size and order subnets for growth, to place ACLs close to the traffic source and NAT at the single Internet boundary, and to compare alternatives (router-on-a-stick vs SVIs; OSPF vs EIGRP vs static routing; WPA2-Personal vs Enterprise).
K6	**Engineering Practice**	Knowledge of accepted engineering practice/technology in the field — e.g. simulation tools, CLI configuration, testing practices. e.g. your use of Cisco Packet Tracer, IOS CLI commands, consistent device naming, configuration documentation, or a verification routine you followed.	Applies? ☒ **Yes** ☐ Partially ☐ No **How:** The work uses Cisco Packet Tracer with real IOS CLI; consistent device names (R-, DSW-, ASW-, SRV-) and interface descriptions; SSH-only management, Syslog/NTP and TFTP backups; and a planned verification routine (show ip ospf neighbor, show ip route, show vlan brief, show ip nat translations, ACL counters, ping/traceroute test matrix) to be recorded in the final report.

#	Knowledge Area	Definition (BAETE) & Guiding Question	Your Response
K7	**Engineering and Society**	Understanding of engineering's role in society: ethics, professional responsibility, and social/economic/environmental impact. e.g. how your ACL policy or IoT automation protects public-safety or resident data, or why your smart city/town/region scenario's real-world context matters to its users.	Applies? ☐ Yes ☒ **Partially** ☐ No **How:** Social and ethical aspects are addressed at a conceptual level: segmentation protects confidential student academic and financial records held by the registrar; isolating IoT devices reduces the risk of building-automation misuse; and occupancy/temperature automation illustrates energy saving on campus. These impacts are argued from the scenario rather than measured, so engagement is partial.
K8	**Research Literature**	Engagement with research literature or authoritative technical references in the discipline. e.g. Cisco's official documentation, RFCs (e.g. on VLSM or private addressing), or any other technical reference you consulted beyond the lecture notes.	Applies? ☐ Yes ☒ **Partially** ☐ No **How:** The project draws on authoritative technical references rather than research literature: RFC 1918, 2328, 3849, 4291, 5340 and 5737, IEEE 802.1Q and 802.11, and Cisco IOS and Packet Tracer documentation (see References). No peer-reviewed research papers are reviewed, so engagement is rated partial.

11. Complex Engineering Problem (CEP) Mapping

For each attribute, indicate whether your project design addresses it, and justify your answer with a specific reference to your scenario or class design — not a general definition restated.

#	Attribute	Definition (BAETE) & Guiding Question	Your Response
P1	**Depth of Knowledge Required**	Resolved using in-depth, fundamentals-based engineering/specialist knowledge — not solved by routine or superficial application of a single taught example. e.g. does your design require you to genuinely apply networking fundamentals — addressing, routing, VLAN segmentation, security — in an integrated way, rather than each in isolation?	Addressed? ☒ **Yes** ☐ Partially ☐ Not Applicable **Justification:** The network works only if addressing, routing, VLANs and security are designed together: OSPF must advertise every SVI subnet produced by the VLSM plan; ACL wildcard masks depend on subnet boundaries (VLANs 98/99 were placed in one aligned /27 so a single entry covers both); NAT exclusions depend on which subnets are IoT and management; and DHCP relay must pass the inbound ACLs. This needs fundamentals-based reasoning (K3–K5) rather than repeating one taught example.
P2	**Range of Conflicting Requirements**	Involves wide-ranging or conflicting technical and other issues that must be balanced against each other. e.g. does your scenario force trade-offs — addressing efficiency vs. scalability, security restrictiveness vs. usability, wireless coverage vs. security?	Addressed? ☒ **Yes** ☐ Partially ☐ Not Applicable **Justification:** Conflicts to be balanced: (i) address efficiency vs summarisation — a compact VLSM plan was chosen over block-aligned ranges that would summarise per building, accepting a single OSPF area; (ii) security vs usability — students are blocked from faculty hosts, yet replies to faculty-initiated sessions are allowed with “established”; (iii) convenience vs security — WPA2-PSK is simpler but weaker than 802.1X; (iv) redundancy vs complexity — the triangle adds a backup path at the cost of an extra link and adjacency.
P3	**Depth of Analysis Required**	Has no single obvious solution; requires abstract thinking and originality to formulate a suitable design. e.g. did you have to weigh multiple possible topologies, addressing schemes, or routing protocols before settling on one, rather than there being one “correct” answer?	Addressed? ☒ **Yes** ☐ Partially ☐ Not Applicable **Justification:** There is no single correct design. Alternatives considered include one core switch with router-on-a-stick, one multilayer switch per zone, EIGRP or static routing, and one management VLAN spanning both blocks. The two-block design with routed links, OSPF and per-block management VLANs was chosen by comparing scalability, fault tolerance, spanning-tree domain size and feasibility in Packet Tracer; choices may still be revised after testing.

#	Attribute	Definition (BAETE) & Guiding Question	Your Response
P4	**Familiarity of Issues**	Involves issues that are infrequently encountered, rather than routine, well-rehearsed problems. e.g. does your ACL policy or IoT automation rule address a scenario-specific condition that is not simply the standard lab-manual example?	Addressed? ☐ Yes ☒ **Partially** ☐ Not Applicable **Justification:** Most techniques (VLANs, OSPF, NAT) are standard lab topics. Some issues are scenario-specific: allowing replies to faculty-initiated sessions while blocking student-initiated access, confining IoT devices to DHCP and their controller across a routed core, excluding IoT/management subnets from PAT, and designing a hysteresis rule. These extend familiar techniques rather than being genuinely unfamiliar problems, so the attribute is rated partial.
P5	**Extent of Applicable Codes**	Goes beyond standard codes of practice — typically most relevant to safety-critical or regulated engineering domains. Often Not Applicable at this project scale — state clearly if no formal codes of practice apply to your scenario, or note any addressing/security standard you chose to follow voluntarily (e.g. RFC 1918 private addressing).	Addressed? ☐ Yes ☐ Partially ☒ **Not Applicable** **Justification:** No mandatory codes of practice or regulations apply to a student simulation, and no regulatory compliance is claimed. The design voluntarily follows technical standards as good practice — RFC 1918 private addressing, RFC 3849 and RFC 5737 documentation prefixes, IEEE 802.1Q tagging and IEEE 802.11 WPA2/AES security — but voluntary adherence is not treated as going beyond codes of practice.
P6	**Extent of Stakeholder Involvement**	Involves diverse groups of stakeholders with widely varying needs and requirements. e.g. does your smart city/town/region scenario need to serve residents, public-safety staff, and network administrators, each with different access needs?	Addressed? ☒ **Yes** ☐ Partially ☐ Not Applicable **Justification:** The scenario defines stakeholder groups with conflicting needs: students (broad access, Wi-Fi), registrar/administration (confidentiality), faculty (teaching and administrative services), researchers (IPv6), facilities staff (dependable automation), IT administrators (secure management) and external portal users. Their needs directly shape the VLAN, ACL and NAT design. These are modelled personas; no real university stakeholders have been consulted.
P7	**Interdependence**	Is a high-level problem made up of many interdependent component parts or sub-problems. e.g. do your routing, VLANs, addressing, security, wireless, and IoT layers all have to work together correctly for the network to function — a change in one affecting the others?	Addressed? ☒ **Yes** ☐ Partially ☐ Not Applicable **Justification:** The layers are tightly coupled: a VLAN or subnet change needs matching SVI, DHCP pool, OSPF advertisement, ACL wildcard and NAT-ACL updates; an over-broad ACL can break DHCP relay or IoT registration; IoT automation depends on routing between VLAN 70 and VLAN 60 across L3; and failover depends on OSPF costs. A fault in one layer therefore affects others, which is why an integrated test matrix is planned.

12. Complex Engineering Activity (CEA) Mapping

For each attribute, indicate whether your project work (design, build, testing, and reporting process) addresses it, and justify your answer.

#	Attribute	Definition (BAETE) & Guiding Question	Your Response
A1	**Range of Resources**	Involves the use of diverse resources — people, time, information, and technologies. e.g. Cisco Packet Tracer, IOS command references, instructor consultation during the Lab 11 design session, or peer discussion used during development.	Addressed? ☒ **Yes** ☐ Partially ☐ Not Applicable **Justification:** Planned resources: Cisco Packet Tracer [version to be confirmed], Cisco IOS configuration guides, relevant RFCs and IEEE standards, lecture notes and the lab manual, consultation with the course instructor in the scheduled design session (Lab 11, as referenced in the brief), peer discussion, and a six-week schedule (Section 9). These resources are planned; no consultation has yet taken place.

#	Attribute	Definition (BAETE) & Guiding Question	Your Response
A2	**Level of Interaction**	Requires resolving significant problems arising from interactions between wide-ranging or conflicting technical issues. e.g. resolving a conflict between your VLAN/addressing plan and your routing configuration, or between your ACL policy and legitimate IoT/wireless traffic.	Addressed? ☒ **Yes** ☐ Partially ☐ Not Applicable **Justification:** The build is expected to require resolving interactions between layers: inbound ACLs on SVIs must not block DHCP relay or IoT registration; the IoT exclusion in NAT-USERS must agree with IOT-PROTECT; passive OSPF interfaces must still advertise SVI subnets; and OSPFv3 support on the 3650 must be confirmed, with static IPv6 routes as a fallback. These problems are anticipated; how they are resolved will be documented in the final report.
A3	**Innovation**	Involves creative use of engineering principles and knowledge in novel ways. e.g. the originality of your smart city/town/region scenario, or a design choice (addressing scheme, routing protocol, IoT automation rule) you arrived at rather than copied from a lab example.	Addressed? ☐ Yes ☒ **Partially** ☐ Not Applicable **Justification:** Original elements include the smart-campus adaptation, per-block management VLANs aligned to one /27 for a single ACL entry, the faculty-reply policy, IoT containment combined with NAT exclusion, and hysteresis in the ventilation rule. All are creative combinations of established techniques rather than new engineering methods, so innovation is rated partial.
A4	**Consequences for Society and the Environment**	Has significant consequences in a range of contexts, where predicting and mitigating those consequences is difficult. e.g. your scenario's public-safety, societal, or sustainability motivation (Lab 4/13's smart-city theme), or a safety/reliability consideration in your network design.	Addressed? ☐ Yes ☒ **Partially** ☐ Not Applicable **Justification:** The scenario has a societal and sustainability motivation adapted from the smart-city theme of Labs 4/13: protecting student records, keeping learning services available through redundant routing, and saving energy by switching lights and fans on demand. As a simulation it has no real-world consequences and its impacts cannot be measured; they are discussed qualitatively, so the attribute is rated partial.
A5	**Familiarity**	Extends beyond previous experience, requiring you to apply principles-based approaches to a new or unfamiliar situation. e.g. does your scenario or a specific technique (e.g. IPv6 addressing, WLC configuration, IoT scripting) go beyond what was directly demonstrated in the lab manual?	Addressed? ☐ Yes ☒ **Partially** ☐ Not Applicable **Justification:** Several elements are likely to go beyond what the lab exercises demonstrated directly: dual-stack IPv6 with OSPFv3 on multilayer switches, a redundant OSPF triangle with failover testing, static NAT combined with an inbound edge ACL, and IoT server rules across a routed network. The exact extent depends on the lab-manual content and should be confirmed with the instructor, so it is rated partial.

13. Student Declaration

I declare that the scenario and design described in this proposal are my own original work, and that I understand the Comprehensive Networking Project brief's required network architecture and deliverables.

14. Instructor Review

Decision: ☐ Approved ☐ Revisions Required ☐ Not Approved	
Instructor Comments:	
Student Signature: Date:	Instructor Signature: Date:

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